

Incident Response Report

Executive Summary:

An actual attack was performed using Metasploit against a vulnerable Samba service on Metasploitable2. The attack resulted in exploiting the vulnerability, which is MITRE ATTACK technique T1210 (Exploitation of Remote Services). The activity was monitored by the SOC using alerts generated by Wazuh and began an incident response process.

Incident Timeline:

At 16:00, the attack took place from the source IP address 192.168.1.102. An alert was triggered by Wazuh that identified potential malicious activities with the Samba service. The alert was verified in TheHive, and it confirmed the malicious exploitation attempt. The threat was confirmed within two hours (MTTD) and responded within four hours (MTTR).

Root Cause Analysis (RCA):

The root cause of this incident can be attributed to the following factors: 1) presence of an exploitable vulnerability in the Samba service; 2) no network segmentation to prevent access to the exploitable resources; 3) configuration of the network to allow any external connection; 4) intrusion detection system not tuned for timely detections.

Response Activities:

The affected VM was isolated from the network in order to avoid any potential lateral movement. The IP of the attacker was blocked via CrowdSec. The process of IP blocking and case creation through the use of the SOAR capability of TheHive was automated.

Metrics and Results:

Mean Time to Detection (MTTD): 2 hours

Mean Time to Response (MTTR): 4 hours

Dwell Time: 6 hours

Recommendations:

Ensure patching activities, implement proper network segmentation, and update your detection logic in Wazuh. Adversary emulation through MITRE Caldera will allow detecting gaps in your defense posture and help mitigate vulnerabilities.